

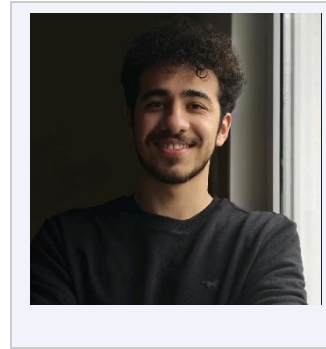
MUHAMMED İZZET DEMİR

Full-Stack Software Engineer | AI & Biometric Systems

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Ankara, Türkiye



ABOUT ME

Full-stack computer engineer graduated from Gazi University, currently building production systems at Papilon A.Ş. that serve 10,000+ users across enterprise clients and defense sectors. I specialize in React/TypeScript frontends, Python/Node.js backends, and RESTful API architecture — delivering scalable, containerized web applications in Agile environments. Beyond full-stack development, I design and ship AI-powered computer vision solutions: face recognition, liveness detection, and OCR systems deployed across web, Android, and iOS. I have published a biometric SDK as an npm package, contributed to a Türk Telekom joint defense project, and consistently deliver across the full development lifecycle from architecture to production.

WORK EXPERIENCE

Papilon A.Ş.

February 2026 – Present

Software Development Specialist | Full-time

- Building production full-stack systems (React + Python/Node.js) for 10,000+ enterprise users across attendance, personnel, and defense sectors.
- Designed and shipped biometric SDKs: on-device face recognition & passive liveness detection (iBeta Level 2 anti-spoofing, zero data leakage).
- Published a biometric verification library as an npm package used in commercial products.
- Contributed to a Türk Telekom joint defense project (IP camera proxy + face analytics API, Maldives deployment).
- Containerized deployments with Docker; Nginx reverse proxy and SSL configuration on Linux infrastructure.
- Agile/Scrum sprint planning, task management, and code reviews via Jira.

Papilon A.Ş.

September 2024 – February 2026

Junior Software Development Specialist | Full-time

- React and TypeScript front-end development; React Native mobile application development.
- Biometric SDK integration and real-time face recognition feature development.

Papilon A.Ş.

September 2023 – August 2024

Software Development | Part-time

- React.js and Redux-based web application development; third-party API integrations.
- Dashboard and component development for internal enterprise tools.

Papilon A.Ş.

August 2023 (1 month)

Software Development Intern

- Internship at Papilon Defense Software Development Department — Web Services API and React.js projects.

T.C. Ministry of Youth and Sports

July 2023 (1 month)

IT Intern | Bilgi İşlem Daire Başkanlığı

- Networking, network security, and IT infrastructure tasks at the Ministry's Information Technology Department.

Anadolu Platformu

2022 – 2023

Front-End Developer | Part-time

- Website maintenance, content management, and responsive UI improvements.

LIVE & COMMERCIAL PROJECTS

- Enterprise Biometric Attendance & HR Platform

• Live — 10,000+ Users

React 18 · Flask · JWT · Leaflet · i18next · PostgreSQL

Multi-role web platform for biometric attendance management, real-time location tracking via Leaflet maps, task scheduling, activity logging, Excel export, and Turkish/English multilingual support. Serving 10,000+ active users in production.
- Agricultural Workforce & Field Operations Dashboard

• Live

React 18 · Material UI · Syncfusion · Recharts · TailwindCSS

Enterprise management panel for the agriculture sector with map-based location tracking, worker assignments, CSV/Excel data export, Swagger API integration, and multi-language support.
- Corporate Personnel Tracking System

• Live

React 18 · Vite · ExcelJS · jsPDF · XLSX

Personnel status tracking dashboard with automated Excel and PDF report generation. Modern Vite build pipeline and clean component architecture for enterprise internal use.
- Defense-Grade Surveillance & Analytics API

• Live — Türk Telekom

Node.js · Express.js · Python · Flask · CORS

IP camera proxy and real-time face detection API developed as part of a joint defense project with Türk Telekom. Handles Digest authentication for IP cameras and logs detection events in real time.
- Biometric Mobile Identity Verification App

• In Development

React Native 0.72 · ML Kit · ONNX Runtime Mobile · Vision Camera

Cross-platform mobile application with on-device face detection, real-time OCR text recognition, and ONNX model inference. Features SSL pinning for security and smooth Reanimated animations.

AI & BIOMETRIC SYSTEMS

- On-Device Face Recognition & Anti-Spoofing SDK

• Live Production

JavaScript · ONNX Runtime Web · Webpack

100% client-side SDK using a NIST FRVT-listed model. Delivers iBeta Level 2 anti-spoofing (print/video/3D mask/deepfake), 512-dim face embeddings, pose estimation, and age/gender analysis. Zero data leaves the browser.
- Real-Time Passive Liveness Detection API

Python · FastAPI · ONNX Runtime · YOLOv5 · PyTorch

REST API classifying frames as real / fake / unknown / blurry using MiniFASNetV2SE anti-spoofing model. Includes Laplacian blur filtering, temporal smoothing across multiple frames, and webcam/video file support.
- Biometric Verification Web Suite

• Published npm Package

React 18 · TensorFlow.js · MediaPipe · Vite

Client-side biometric verification suite published as an npm package. Provides step-by-step liveness confirmation with face angle/pose detection, smile detection, and image quality scoring.
- Enterprise Multi-Camera Face Recognition System

Python · PyQt6 · Flask · YOLOv8 · PostgreSQL · ONNX

Desktop application with REST API backend supporting USB, IP, and RPC cameras. Features YOLOv8 body detection, embedding-based user registration in PostgreSQL, BLE Wallet device integration, and MVC architecture.
- Additional Technical Projects

Android On-Device Liveness Detection SDK

—

Native Android SDK with 5-frame temporal smoothing and adjustable confidence threshold

[Kotlin · CameraX · ONNX Runtime · YOLOv5]

Browser-Based ID Document OCR Engine

— Active —

client-side OCR for identity documents, 6 languages, PWA offline support, 3 engine options

[JavaScript · ONNX Runtime Web · Tesseract.js · OpenCV.js]

Face Detection & Identity Management REST API

—

JWT-secured API with Swagger docs, ONNX inference, and migration-based schema management

[Python · Flask · PostgreSQL · ONNX · JWT · Swagger]

Face Detection Analytics Dashboard

—

Real-time statistics and visualization frontend for the Face Detection API

[React · Vite · Bootstrap · DataTables]

FaceNet Custom Training Pipeline

—

Transfer learning with triplet loss optimization on custom identity datasets

[Python · PyTorch · TensorFlow · Scikit-learn]

TECHNICAL SKILLS

Frontend	React 18, TypeScript, Vite, Next.js, TensorFlow.js, Material UI, TailwindCSS, Bootstrap, Chakra UI, Leaflet, i18next, DataTables
Backend & APIs	Python, FastAPI, Flask, Node.js, Express.js, RESTful APIs, JWT, PostgreSQL, SQLAlchemy, Swagger/Flasgger
AI / Computer Vision	PyTorch, TensorFlow, ONNX Runtime, OpenCV, YOLOv8/YOLOv5, MediaPipe, FaceNet, MTCNN, Liveness Detection, Anti-Spoofing
Mobile	Kotlin (Android), CameraX, Room, WorkManager, Flutter/Dart, React Native, ML Kit, ONNX Runtime Mobile, Vision Camera
DevOps & Tools	Docker (multi-arch), Nginx, Linux, Git, Bitbucket, Postman, Jira, npm package publishing
Methodologies	Agile/Scrum, Sprint Planning, Code Review, REST API Design, MVC Architecture

EDUCATION

Gazi University - Computer Engineering

Bachelor of Science | Graduate | 2.91

Ankara, Türkiye

CERTIFICATES

- 2023 Python for Data Science, AI & Development - IBM
- 2023 WEB 301 - Turkcell
- 2023 WEB 401 - Turkcell
- 2025 Version Control- Coursera
- 2025 Yapay Zeka ve Algoritmalarına Giriş – BTK

LANGUAGES

Turkish - Native | English - Professional